

BEFORE EVENING

An AI-Driven VR Narrative Experience Game

PROCESS BOOK

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3 9 '26

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Chapter 1: Project Overview

Lead: Lu Zhengping



1.1 Project Vision

Before Evening is a VR narrative experience built on a single hypothesis: that the future of interactive storytelling lies in characters who truly listen. Rather than choosing from preset dialogue options, players speak naturally to photorealistic MetaHuman NPCs, each with their own memories, personalities, and understanding of the story. The result is an experience where every conversation is unique, and every playthrough reveals the story differently.

The game is set in a Chinese high school at dusk. The player takes the role of Yi Linlan, a young teacher returning to her former school, only to discover that the girl she once bullied has also returned — now as the school psychologist. What unfolds is a quiet but tense investigation into memory, guilt, and confrontation, told entirely through environmental exploration and natural conversation.

1.2 Core Innovation

MetaHuman Photorealism: Epic Games' MetaHuman framework provides cinematic-quality facial detail, hair, and skin. In VR, proximity to the human face creates emotional weight that stylized characters cannot match.

AI-Driven Dialogue via Convai.AI: Each NPC is connected to an LLM-backed agent with custom personality, memory, and contextual awareness. Players speak; NPCs respond in their own voice with character-appropriate reactions. There are no dialogue trees.

Voice-First VR Interaction: Players hold the right A button to speak, freeing their hands for spatial exploration and object manipulation. This pushes VR away from menu-driven design and toward genuine presence.

1.3 Project Specifications

- **Platform:** Meta Quest 3
- **Engine:** Unreal Engine 5.7.4
- **AI Backend:** Convai.AI
- **Character System:** MetaHuman + Quixel Bridge
- **Demo Length:** 15–20 minutes
- **Genre:** Narrative VR / Investigation
- **Target Frame Rate:** 72 FPS

1.4 Team & Division of Labor

- **Lu Zhengping:** Project Manager. Environment Art (school scene from scratch). VR Interaction Programming (Blueprints for movement, grabbing, triggers).
- **Gong Shengao:** MetaHuman Integration. AI Prompt Engineering. Atmospheric Effects (fog, rain, lighting polish). Final presentation and video production. Narrative Designer. Story background, character writing, mission structure, and AI prompt content.

- **Lin Yang:** 3D Asset Design. Texturing of key props (office furniture, classroom items, posters, narrative-relevant objects).Level Design and player flow. UE5 Blueprint logic. VR input routing, event triggers, state machines.

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Chapter 2: Concept Ideation

Lead: Gong Shengao

2.1 Overview

The VR project explores psychological suspense and emotional confrontation within a school environment. Players engage in a loop of exploration, dialogue with AI-driven MetaHuman NPCs, evidence tracking, and emotional confrontation. Environment, objects, and NPC interactions are designed to reflect memory, guilt, and hidden past actions, making the environment itself a narrative tool.

2.2 MetaHuman Integration

Four distinct NPCs are represented as MetaHumans, each with dynamic facial expressions, natural reactions, and unique memory systems. Player interactions with these NPCs provide dynamic narrative feedback and allow clues to emerge organically, reinforcing immersion. MetaHuman models informed concept decisions, influencing lighting, posture, facial expressions, and spatial narrative flow.

2.3 Inspirations

- The Glory (Netflix): Explores institutional silence, perpetrators rewriting history, and narrative-driven psychological character arcs.
- Silent Hill 2 (Konami): Uses environmental narrative to materialize guilt, trauma, and dual-layer reality.
- P.T. (Hideo Kojima): Demonstrates spatial repetition, micro-changes, and minimalist interaction design to create tension.
- Real Photography: Collaborations for authentic images integrated into the environment to enhance immersion.



2.4 Design Decisions

- Environment as Memory: Spaces evolve across loops reflecting psychological states.
- Objects as Clues: Lighter, photographs, and old files serve both gameplay and narrative purposes.
- MetaHuman-Informed Ideation: Character expressions and postures influenced lighting, object placement, and scene composition.

2.5 Design Insights

- MetaHuman integration allows precise control over emotional cues and narrative flow.
- Subtle environmental cues convey tension without scripted events.
- AI-driven NPC interactions provide feedback loops, informing ongoing concept refinement and pacing.

2.6 Visual and Spatial Exploration

- Concept art emphasizes psychological tension, lighting, and spatial composition.
- Looping corridors, offices, and classrooms reflect both physical and mental states.
- Objects interact with the player not only as gameplay elements but as narrative devices, shaping memory and emotional response.

2.7 Integration Summary

The Concept Ideation phase establishes a strong foundation for the VR narrative by combining MetaHuman-driven NPCs, environmental storytelling, and object-based narrative clues. This integration ensures that players experience tension and immersion organically, and that the environment dynamically communicates both past and present psychological states.

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Chapter 3: Narrative Structure

Lead: Lin Yang

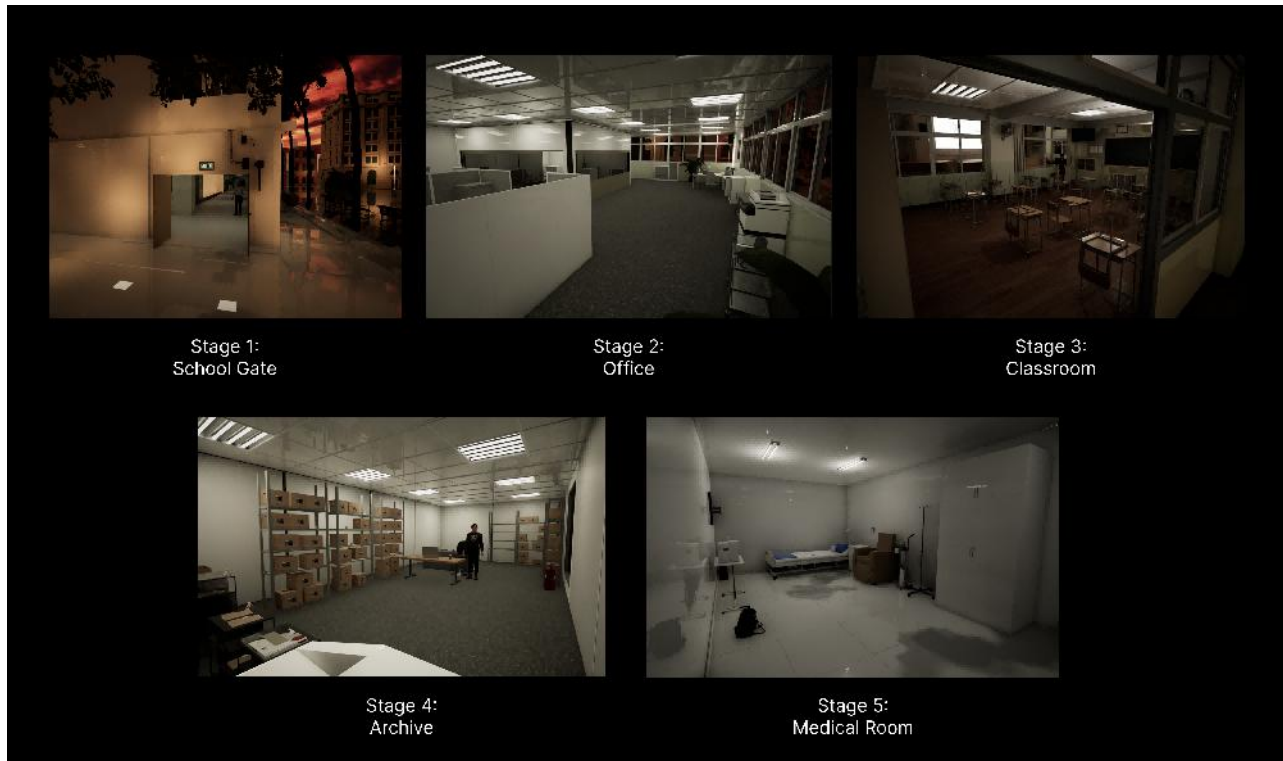
3.1 Story Background

Players control Linlan Yi, who returns to her alma mater as a teacher. Previously a high-achieving student with hidden bullying behavior, she now faces the resurfacing of past actions when a former victim has returned as the school psychologist. The narrative unfolds through five immersive stages, each designed to escalate emotional intensity, deepen psychological engagement, and guide the player through both external investigation and internal reflection. The school environment itself serves as a character, reflecting the protagonist's memory and past deeds, and dynamically responding to player exploration and interaction.

Stage Breakdown

Stage	Scene	Narrative Function	Gameplay Mechanics	NPC
1	School Gate	Introduce campus and protagonist; establish calm baseline	Walking + initial dialogue; environmental introduction	Principal Zhao
2	Office	Discovery of anomalous files; triggers investigation	Object inspection + subtle environmental cues; interactive exploration	Environmental objects
3	Classroom	Revisiting memories; emotional escalation	Dialogue with Teacher Zhou, spatial exploration, memory flashbacks	Teacher Zhou
4	Archive	Transform	Numbered archive	Archive

		suspicion into concrete evidence	search, procedural puzzles, document cross-referencing	Administrator
5	Medical Room	Psychological confrontation; culmination of narrative	Free dialogue with branching options, AI-driven NPC reactions, adaptive emotional feedback	Psychologist He



3.2 Narrative Mechanics

- Environmental Loops: Corridors, classrooms, and offices evolve slightly with each loop, reflecting psychological destabilization and memory resurfacing.
- Fragmented Evidence: Objects such as photographs, files, lighter, and other environmental elements

serve as narrative anchors for reconstructing the protagonist's past actions.

- Dynamic NPC Dialogue: AI-driven memory systems allow NPCs to respond naturally to player input, with subtle variations in tone, facial expression, and choice of words, creating emergent storytelling and adaptive pacing.
- Player Choice and Reflection: While core story beats remain fixed, the player's investigative choices and interactions influence narrative rhythm and emotional tension, allowing for a personalized experience.

3.3 Player Journey

(originally planned, scaled down in final build)

1. Initial Calm: Introduction to protagonist and school aesthetics, establishing baseline mood and environmental familiarity.
2. Emerging Tension: Office anomalies and old files spark curiosity and unease, guiding the player towards investigation.
3. Emotional Nostalgia: Classroom revisit triggers memory flashbacks and emotional resonance through dialogue and environmental cues.
4. Systemic Pressure: Archive exploration imposes procedural logic, combining evidence tracking with narrative discovery.
5. Climactic Confrontation: In the medical room, interactions with He Qinyue integrate previously discovered evidence and memories, resulting in psychological and emotional resolution.

3.4 Emotional & Narrative Integration

- Each stage is carefully designed to gradually escalate tension, blending environmental cues with AI-driven character interactions.
- Loops, spatial manipulation, and subtle environmental changes reinforce the protagonist's memory, guilt, and psychological state.
- Player actions dynamically influence NPC responses and narrative pacing, ensuring emergent storytelling without breaking core narrative structure.

- The design ensures that both exploration and confrontation feel organically connected, delivering a cohesive, immersive narrative experience that merges psychological suspense with interactive investigation.

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Chapter 4: Gameplay Design

Lead: Gong Shengao

4.1 Core Gameplay Loop

The gameplay loop is intentionally simple: Explore → Encounter → Converse → Reflect → Move on. Players move through the school, encounter NPCs, hold the right A button to speak, and continue. There are no quests, no checkpoints, no UI prompts. The school's architecture and the NPCs themselves provide all the guidance the player needs.

4.2 VR Controller Mapping (Quest 3)

The control scheme is built on one principle: separate movement from rotation, and never compete with the headset.

- **Left thumbstick (forward):** Walk forward
- **Left thumbstick (left/right):** Strafe left/right
- **Right thumbstick (left/right):** Snap rotation
- **Both thumbsticks forward simultaneously:** Sprint
- **Right A button (hold):** Talk to NPC (activates Convai voice channel)
- **Right grip + trigger:** Grab and examine object
- **Left menu button:** Open menu

CONTROLLER MAPPING (QUEST 3)



4.3 Design Rationale

- **Strafe added in iteration 2:** Forward-only movement felt restrictive; players wanted to sidestep around objects and NPCs.
- **Snap rotation replaces smooth rotation:** Smooth controller rotation conflicted with natural head-turning, causing motion sickness.
- **Dual-stick sprint emerged organically:** Players were already pushing both sticks forward instinctively, so we mapped that gesture to faster movement.
- **Hold-to-talk over toggle:** Hold-to-talk gives players precise control over when NPCs are listening, avoiding accidental capture of side conversations.

4.4 Interaction Systems

- **Locomotion:** Walking, strafing, and sprinting through the environment.

- **Object Interaction:** Players can grab, rotate, and examine physical objects (books, papers, photos, props). Object physics are simulated; objects can be placed back, dropped, or carried.
- **NPC Interaction:** The voice channel. Hold A, speak, release. The NPC's MetaHuman face responds with synchronized lip-sync, gesture, and gaze. Players can also show NPCs an object they're holding, and the NPC will respond contextually to it.

4.5 Player Guidance

UI markers, quest arrows, and waypoint systems were deliberately removed. Guidance is delivered through architecture, NPC dialogue, sound, and lighting:

- **Architecture:** Lit corridors lead somewhere; dark ones don't.
- **NPC dialogue:** Characters can verbally point players toward locations or other NPCs.
- **Sound:** Rain audio is louder near windows and exterior doors, helping players orient themselves.
- **Light:** Warm interior lights mark active spaces; cooler tones mark transitional ones.

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Chapter 5: Scene Design

Lead: Lu Zhengping

5.1 Design Philosophy

Lu Zhengping built the school environment from scratch over the course of the project. Working alone on architecture allowed for a unified visual language across all spaces.. The goal was not photorealism — it was familiarity. Anyone who attended a Chinese high school should recognize this place instantly: the proportions, the layout, the way the corridors echo, the way the windows are set too high to see out of comfortably.

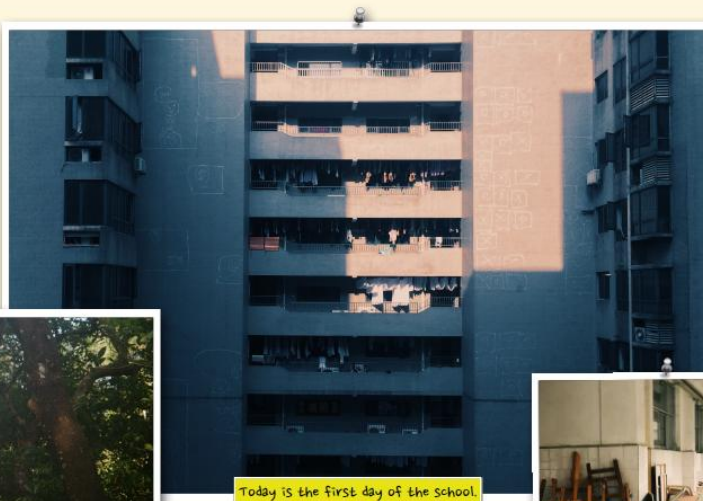
This recognition is the foundation of the game's emotional weight. The story is about returning somewhere you used to belong, and the architecture had to carry that feeling before any character spoke a word.

5.2 Reference & Floor Plan

- **Reference gathering:** Two weeks of photographic research focused on Chinese senior secondary schools — floor plans, ceiling heights, window placements, and faculty office layouts.
- **2D floor plan:** Drafted to cover main entrance, corridor system, faculty office cluster, psychologist's office, classrooms, stairwells, and storage.
- **Compactness:** Plan is deliberately small — full traversal in under five minutes — but folds back on itself to feel larger.

There is a story behind every photo.

PHOTOGRAPHY BY SOPHIA & SEBASTIAN



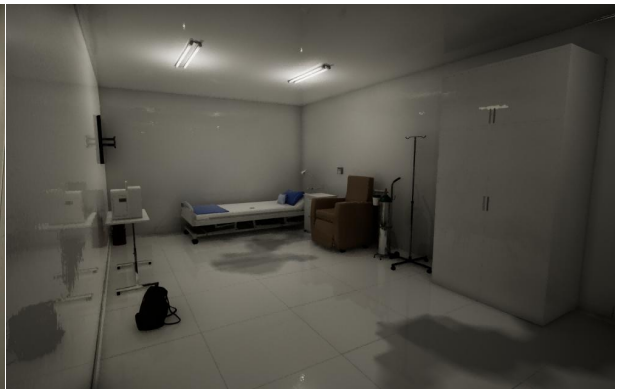
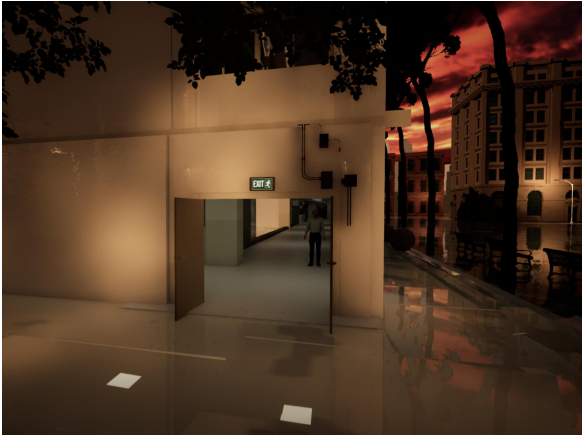
Examples of in-game found photographs that players can pick up.

5.3 Horizontal Design & Spatial Hierarchy

The school is one floors. Vertical movement is intentional and narrative:

- **Ground floor:** Public, lit. Classrooms and common areas.

This vertical separation reinforces the narrative. The bright side is where Yi Linlan can pretend nothing is wrong. The next side is where the conversations she has been avoiding take place.



5.4 Environment

After Round 2 user testing revealed that the empty exterior broke immersion when players looked through windows, a three-layer background system was built:

- **Near layer (5–15m):** A neighboring school building with detailed window frames, brickwork, and a few visible interior shapes.



- **Mid layer (15–30m):** Lower-detail residential buildings, trees, and a parking area. Provides depth and urban context.



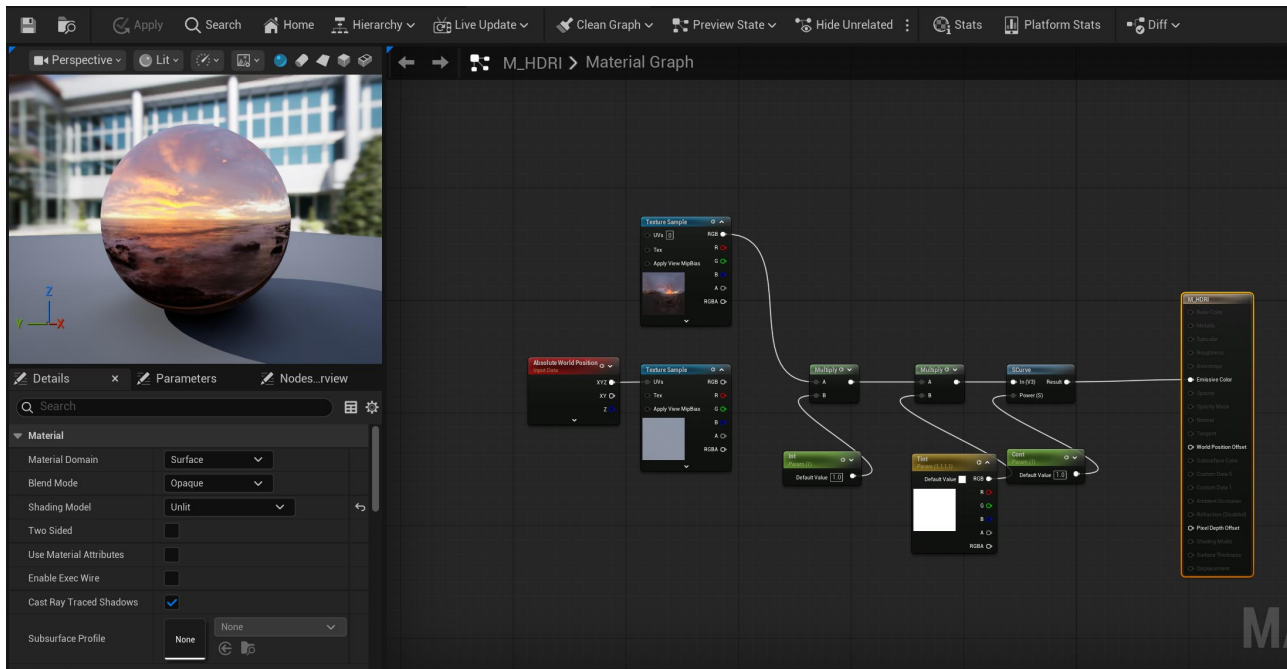
- **Far layer (30m+):** Painted skyline cards, distant city silhouettes, and the skybox itself. No real geometry.



5.5 Skybox & Atmosphere

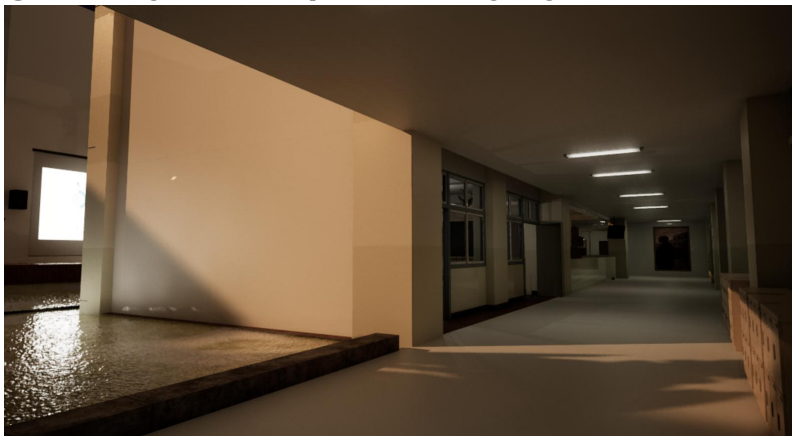
The game's title is the visual brief: Before Evening. The skybox is a custom dusk gradient — warm amber at the horizon transitioning to deep blue overhead, with low-hanging clouds catching the last light. The sun is positioned low and to the west, casting long horizontal shadows through the corridor windows.

Dusk carries an inherent sense of ending, of running out of time, of decisions deferred until it is almost too late. This single design choice does more emotional work than any other element in the scene.



5.6 Lighting Strategy

- **Directional sunlight:** Low angle, warm temperature, casting long shadows.



- **Window light shafts:** Volumetric fog catching the rays.



- **Practical interior lights:** Fluorescent overhead in corridors, warmer desk lamps in offices.



- **Negative space lighting:** Dark stairwells and unlit rooms create natural visual flow toward lit destinations.



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Chapter 6: Character Art

Lead: Cai Qingling

6.1 Protagonist

- Linlan Yi / 伊林岚: Linlan Yi embodies the 'golden child' archetype. Visually flawless, she conceals past guilt, envy, and bullying behavior, creating tension between outward perfection and internal conflict.
- Narrative Function: Player is both investigator and subject; environmental cues, objects, and interactions reveal her inner turmoil, forcing the player to confront moral and emotional complexity.
- Design Approach: Concept art generated using Google Gemini; emphasis on facial expressions, posture, and subtle gestures to reflect psychological tension. Lighting and shadow play are strategically used to highlight internal conflict during interactions and exploration.



Character

Evelyn

伊林岚

She is the quintessential "golden child." In middle school, hidden behind stellar grades and flawless obedience, her envy of a brighter peer drove her to orchestrate covert bullying.

Using a lighter to burn the victim's belongings, she terrorized them in secret while teachers, blinded by her perfect facade, ignored all pleas for help.

Masterful at self-rationalization, she has rewritten her own memories, never once seeing herself as the villain.

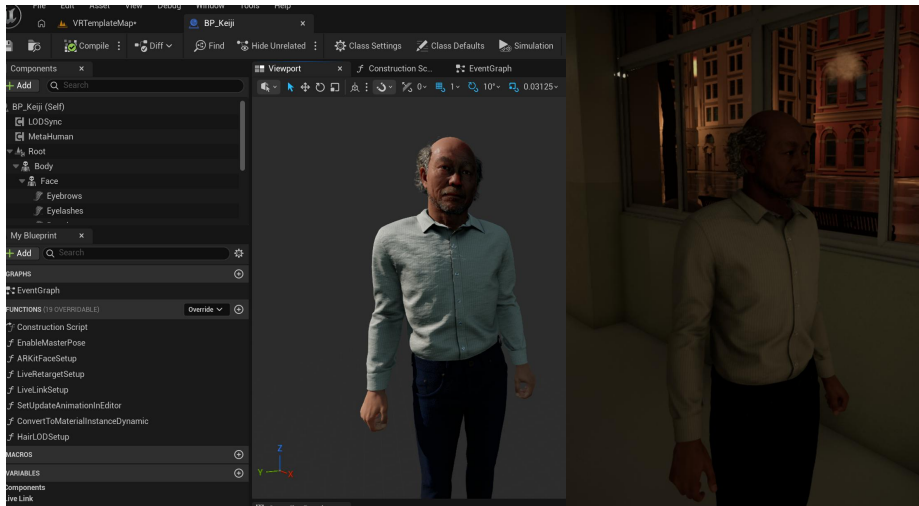
Returning to her alma mater as an "Outstanding Educator," her new title seamlessly continues her old disguise: forever respectable, forever righteous, and never the abuser.

CONCEPT ART GENERATED BY Google Gemini

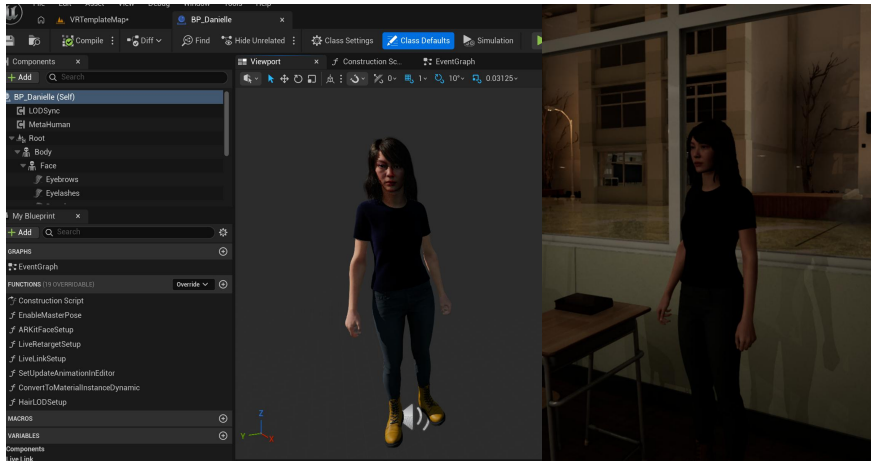
6.2 Supporting Characters

Four distinct NPCs, each driven by its own MetaHuman system with individual memory and dynamic responses, designed to interact meaningfully with the player.

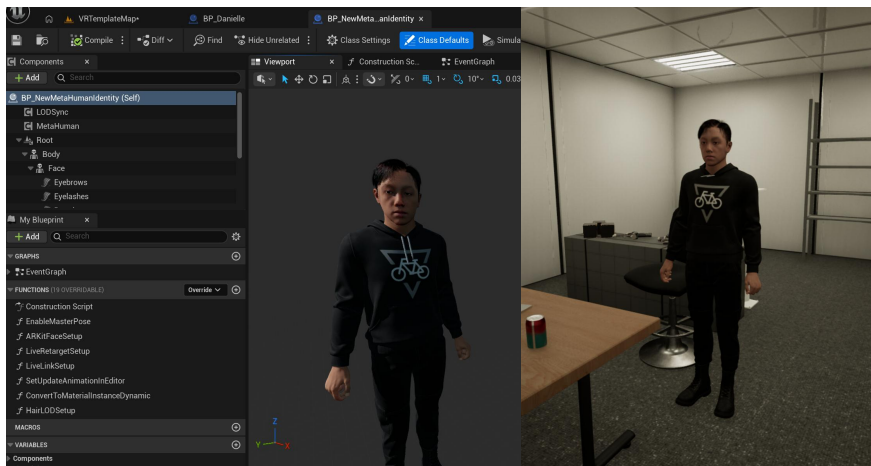
1. Zhao Qiming / 赵启明 – Principal, establishes the official welcoming tone; demeanor calm and composed to contrast the protagonist's rising tension. Concept art emphasizes authority, upright posture, and steady eye contact.



2. Zhou Mingzhi / 周明志 – Teacher, emotional bridge; connects protagonist to past memories, guiding narrative transitions. Visual design includes subtle aging cues, reflective facial expressions, and movement patterns that evoke nostalgia.

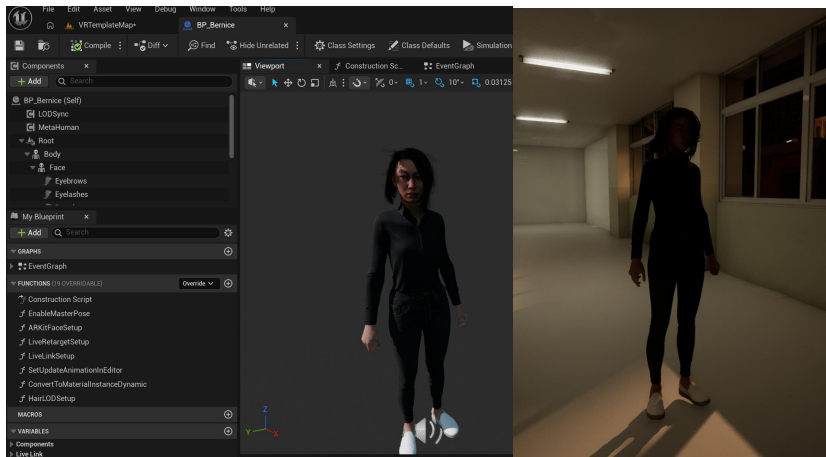


3. Archive Administrator / 档案管理员 – Procedural, enigmatic; enforces rules and system logic. Design emphasizes minimal expressions, rigid postures, and environmental control cues to reinforce the institutional atmosphere.



4. He Qinyue / 何沁月 – Psychologist, final confrontation; strategic, cold, and observant. Character art

leverages sharp lighting, precise gaze direction, and micro-expressions to deliver psychological pressure.



6.3 Character Art Approach

- Concept Development: Iterative creation of facial expressions, posture, and subtle gestures to convey emotional states. Concept art aligned with gameplay triggers and environmental context.
- Environmental Integration: Characters influence and respond to lighting, spatial cues, and interactive objects. MetaHuman-driven expressions allow real-time reaction to player actions, enhancing immersion.
- Interaction Design: Each NPC's memory system determines dynamic responses, dialogue variations, and emotional cues. Player choices alter subtle behaviors and facial expressions, reinforcing emergent storytelling.

6.4 Visual & Narrative Integration

- MetaHuman-driven characters provide responsive, lifelike reactions to player exploration and dialogue.
- Dynamic interactions ensure that story progression is linked to both exploration and emotional engagement.
- Environmental cues, visual details, and character positioning are carefully coordinated to maintain narrative coherence and psychological tension.

6.5 Character Design Reflections

- Focused on how each character's visual design and behavioral traits contribute directly to narrative and emotional impact.
- Iterative adjustments in posture, micro-expressions, and lighting ensure coherence with emotional pacing and player interactions.
- Dynamic reactions and memory-informed behaviors allow players to experience a story that responds to both exploration and emotional engagement.
- Emphasizes that characters are active narrative agents, not static visuals, reinforcing immersion and psychological depth.

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Chapter 7: Animation

Lead: Gong Shengao

7.1 Facial Animation — The MetaHuman + Convai Bridge

The most technically demanding integration in the project was getting Convai.AI's audio output to drive MetaHuman's facial animation system in real time. The pipeline:

- **Step 1:** Player speaks → Convai Speech-to-Text
- **Step 2:** LLM generates response → Convai Text-to-Speech
- **Step 3:** Audio stream → MetaHuman Audio2Face
- **Step 4:** Facial blendshapes → Synchronized playback

Each step adds latency. Each step also has to handshake with the next step in a way that does not break under network jitter or processing spikes. This pipeline was rebuilt three times before it worked reliably

7.2 Body Animation

Body animation is intentionally minimal. Each NPC has:

- **Idle loop:** Breathing, weight shifting, occasional glances.
- **Speaking loop:** Light gesture variation tied to speech energy.
- **Listening pose:** Subtle, attentive, suggesting cognitive processing.
- **Contextual reactions:** Nodding, tilting head, looking at offered objects.

No custom motion capture was performed. All body motion comes from MetaHuman's stock library, blended via Animation Blueprints.



Changed quite a bit over the years.



no one bothered to tell me it was you.



If you want to know more about the woman who removed file five, I

7.3 Prompt Engineering as Animation Direction

A surprising finding: a large portion of "animation work" was actually prompt engineering. The prompt determines what the character says, and what they say determines how they move (because Audio2Face derives gesture intensity from speech energy).

- **"You are tired and resigned"**: Produces softer audio, smaller gestures, and a more inward-facing posture.
- **"You are nervous and have something to hide"**: Produces faster speech, more frequent micro-expressions, and shifting gaze.

Roughly 60 hours were spent iterating on prompts to get each character's animation feel right. This was the single most underestimated time sink in the project.

7.4 Technical Challenges

UE5 version incompatibility: A mid-project UE5 update broke the Convai plugin. Engine version was locked thereafter.

- **Skinning artifacts**: Some MetaHuman wardrobe pieces did not deform cleanly during animation. Fixed via manual weight painting.
- **Lip-sync drift**: Audio and facial animation occasionally desynced during long responses. Mitigated by chunking responses into shorter segments.

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Chapter 8: Development

Lead: Lu Zhengping

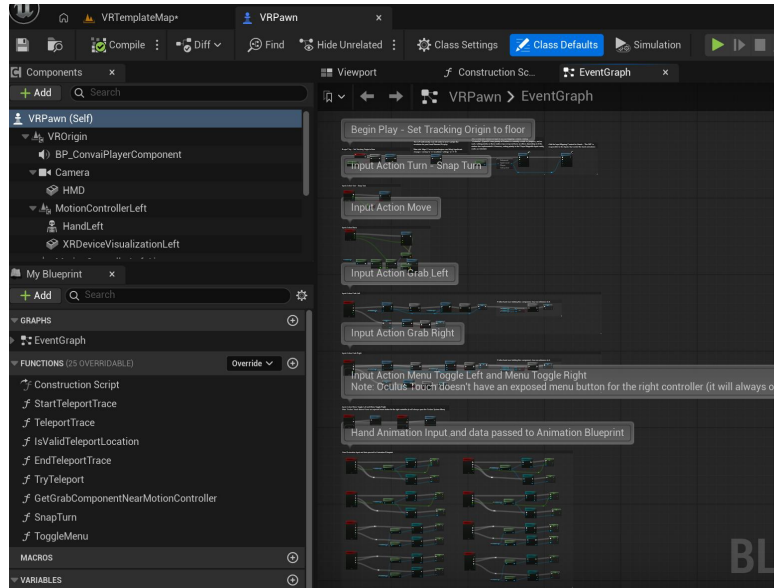
8.1 Development Stack

- **Engine:** Unreal Engine 5.7.4 (locked)
- **VR Platform:** Meta Quest 3 (Link mode for development)
- **AI Backend:** Convai.AI
- **Character System:** MetaHuman + Quixel Bridge
- **3D Software:** Blender, Substance Painter
- **Voice Cloning:** Convai voice module with team member voice samples

8.2 VR Interaction Development — Lu Zhengping

The entire VR interaction layer — locomotion, object interaction, and the talk-to-NPC trigger — was built in Blueprints rather than C++ for iteration speed.

- **Locomotion blueprint:** Three major rewrites. Final version handles independent translation (forward + strafe), independent rotation (snap turn), dual-stick sprint detection, comfort vignette during fast motion, capsule collision, and ground stick.
- **Object interaction:** Overlap volumes around each interactive prop. When the player's hand enters the volume and presses grip, the object attaches to the hand controller's transform. Releasing grip returns the object to physics simulation.
- **Talk trigger:** Hold-to-talk system bound to right A. While held, the microphone is opened and audio is streamed to Convai. On release, the response is fetched and played through the NPC's MetaHuman.



8.3 MetaHuman & AI Integration — Gong Shengao

- **MetaHuman pipeline:** Concept brief → MetaHuman Creator → wardrobe → UE5 import via Quixel Bridge → Convai binding.
- **Convai integration:** Each MetaHuman is bound to a Convai agent with custom personality prompt, memory bank, and voice ID.
- **Atmospheric effects:** Volumetric fog tuning, rain particle system, and lighting polish pass. Rain density was carefully balanced — too dense dropped framerate, too sparse went unnoticed.

8.4 Level & Blueprint Systems — Yang Yulai

The level Blueprint manages:

- **Initial spawn and orientation:** Player position and view direction at start.
- **NPC spawning and despawning:** Maintains the one-MetaHuman-visible limit.
- **Door states:** Locked, unlocked, opened — driven by narrative progress.
- **Trigger volumes:** Cue ambient audio or lighting changes when player enters.
- **State machine:** Tracks which NPCs the player has spoken to.

VR input routing was also handled here — making sure controller inputs go to the right system (locomotion, interaction, dialogue) without conflict.

8.5 3D Asset Production — Lin Yang

Every prop in the game that wasn't part of the architectural shell was produced or sourced individually. This includes:

- **Office furniture:** Desks, chairs, filing cabinets, computers.
- **Classroom props:** Desks, blackboards, posters, books.
- **Interactive objects:** Papers, photos, small personal items.
- **Wall decorations:** Posters and printed signage.

Each prop went through: concept reference → modeling in Blender → UV unwrapping → texturing in Substance Painter → import to UE5 → collision setup → placement. Custom posters and signage were designed based on Cai Qingling's narrative document, so the visible text in the world is consistent with the story.

8.6 Narrative Implementation — Cai Qingling

Narrative implementation continued throughout development. Convai prompts for all four NPCs were written and revised across three iteration rounds based on what players were actually saying during testing. All environmental text — posters, notes, signage — was written to maintain consistency with the story.

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Chapter 9: Sound Design

Lead: Gong Shengao

9.1 Implemented Audio

- **Rain ambience:** Spatially attenuated rain audio plays continuously. Louder near windows, softer in interior rooms. The most consistent audio cue in the game, contributing significantly to the dusk atmosphere.
- **NPC voice (Convai.AI):** All NPC speech is generated through Convai's TTS module. The voices of team members were cloned to give each NPC a distinct, human timbre — a deliberate choice over generic synthetic voices, making the characters feel specific rather than templated.
- **Demo trailer music:** The video presented at the exhibition uses licensed background music. Not present in the playable build.

9.2 Planned but Not Shipped

- **Footstep audio:** With surface variation (tile, carpet, wood).
- **Ambient school sounds:** Distant chatter, HVAC hum, fluorescent buzz.
- **Object interaction sounds:** Paper rustle, drawer slide, object pickup.
- **Dynamic music layer:** Responding to narrative tension.

These were cut due to time pressure in the final two weeks. Their absence is the single most noticeable gap in the shipped build's audio design.

9.3 Audio Lessons

VR audio is more important than flatscreen audio. With no screen edges to remind players they are in a game, missing audio breaks immersion harder than a missing visual element. If a v2 is shipped, the entire ambient and footstep layer must come first, before any new visual content.

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Chapter 10: User Testing & Iteration

Lead: Lin Yang

Three rounds of structured user testing were conducted during development. Each round informed concrete changes to the build..

10.1 Round 1 — Locomotion Validation

- **Build state:** Forward movement and rotation only. Grayboxed environment. No NPCs.
- **Critical feedback:** "Can't sidestep — it feels like I'm on rails." "When I turn my head and the controller turns at the same time, I get dizzy." Two testers had to remove the headset within 10 minutes.
- **Diagnosis:** Smooth rotation from the right stick was conflicting with natural head rotation, creating a double-input vestibular mismatch — a textbook cause of VR motion sickness.
- **Iteration (Lu Zhengping):** Added strafe to the left stick. Replaced smooth rotation with snap rotation. Added comfort vignette during locomotion. Tuned acceleration curves.
- **Outcome:** Round 2 testers reported no comparable motion sickness.

10.2 Round 2 — Immersion & Environment

- **Build state:** Updated locomotion, near-complete interior, basic object interaction. Exterior environment minimal (empty void outside windows).
- **Critical feedback:** "Everything inside feels real — but when I look out the window, it ruins it." "I noticed the school is floating in nothing."
- **Diagnosis:** Player attention naturally drifts to windows in any indoor environment. Exterior detail had been underbudgeted.
- **Iteration (Lu Zhengping):** Built the three-layer LOD exterior described in Chapter 5. Added trees, neighboring buildings, parking, and atmospheric fog. Adjusted skybox to match interior lighting temperature.
- **Outcome:** Window views became an asset rather than a liability. Several Round 3 testers spent extended time just looking outside.

10.3 Round 3 — Full Experience

- **Build state:** All systems integrated. MetaHuman NPCs active. Convai dialogue functional.
- **Positive feedback:** "The school feels like a real place." "I forgot I was in a game when I picked up the paper." "Talking to her felt strange — in a good way. Like she was actually thinking."
- **Critical feedback:** "There's a 3–4 second gap between when I stop talking and when she answers." This was the dominant criticism. Convai latency, while technically expected, broke conversational flow. Body language was also called "a little stiff."
- **Iteration attempts (Gong Shengao):** Investigated streaming response (partial fix). Added "thinking" idle animations to fill silence. Tuned prompts to produce shorter, faster responses.
- **Outcome:** Latency reduced from ~5 seconds to ~3 seconds, but not eliminated. This is the project's most important unsolved problem and the top priority for any future v2.

10.4 Testing Lessons

- VR cannot be designed on a flatscreen. Issues invisible in the editor are obvious within thirty seconds in the headset.
- Tester comfort feedback should be treated as a hard requirement, not a preference.
- Every iteration round should introduce only one major change, so feedback can be attributed cleanly.

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Chapter 11: Exhibition

Lead: Lin Yang

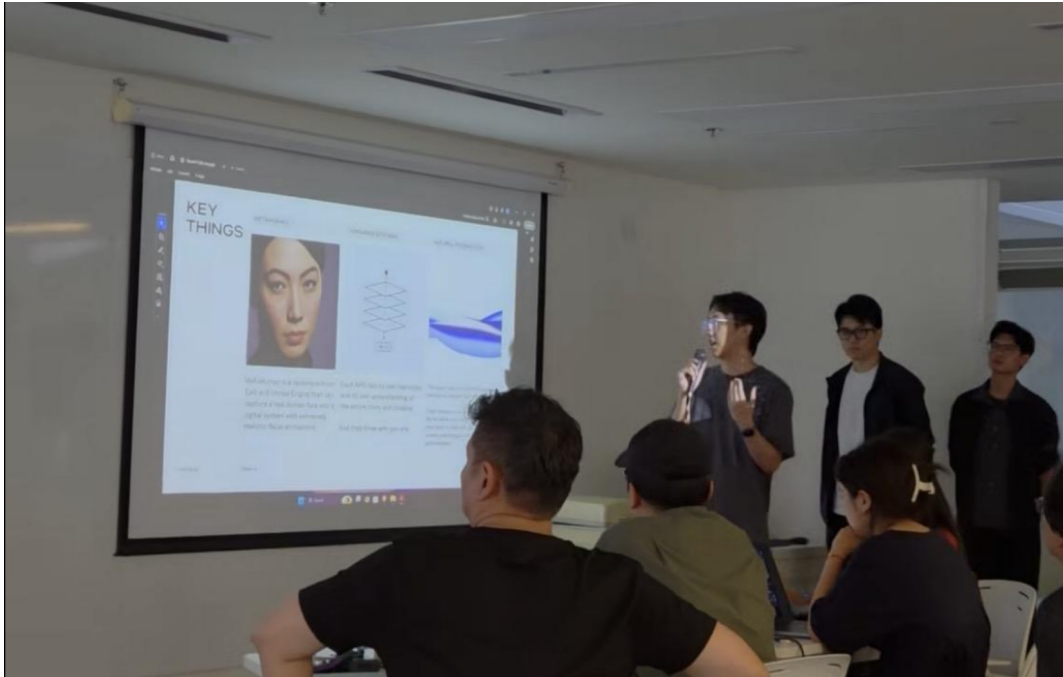
Presentation as translation: turning a live VR build into a clear visual and narrative format for the audience.

11.1 Exhibition Objectives

The final presentation introduced Before Evening as a VR narrative project about school bullying, memory, guilt, and revenge. In the current story, Yi Linlan returns to her old school as a teacher and discovers that the girl she once bullied has returned as the school psychologist. Our exhibition therefore had to communicate not only the plot premise, but also the emotional tension of returning to a space that still carries the past. Just as importantly, it had to explain the project's shift from a broader horror structure to a more focused MetaHuman- and AI-driven narrative built around four NPCs, clue discovery, and spoken interaction.

11.2 Presentation Format and Flow

The presentation on 21 April 2026 used slides, live speaking, and a showcase video. The spoken section established the design pivot, the AI and MetaHuman systems, and the basic interaction logic. The video then carried the rest of the presentation by showing atmosphere, spatial transitions, clue materials, and character encounters. This format worked as a design choice: rather than relying on a fragile live VR demo, it translated the project into a more legible sequence of ideas, images, and narrative cues.



11.3 Video Demonstration Strategy

The video was structured as an introduction to the current game language. It begins with title and atmosphere shots, moves through the campus and corridor spaces, introduces clue materials such as the injury record and wall posters, and then gives extended time to MetaHuman conversations. This made the project's current priorities visible: environmental storytelling, AI dialogue, and character perspective. At the same time, the video still reads more like a curated narrative showcase than a full gameplay demonstration, because tone, clues, and long conversations are more visible than active progression.



11.4 Intended Audience Experience

The exhibition was designed to let the audience read the school as a layered visual system rather than a neutral setting. The office, classroom, infirmary, corridor, and outdoor spaces all hold traces of the same event, while the four NPCs offer different memories of it. The material language of the project supports this reading: the mental-health poster reflects institutional care, the injury report makes violence documentary, the Future Campus poster suggests official distance, and the handwritten red message introduces a more intrusive memory logic. Together, these assets turn the environment into a carrier of accusation, not just background decoration.

noted that the project's direction holds significant ductility and expansion potential, identifying the MetaHuman + AI dialogue pipeline as an architecture that could scale beyond a student demo into a longer-form narrative game. This endorsement of the technical direction — separate from any specific implementation in the current build — was the most important critical signal received during the exhibition. At the same time, they identified the main weaknesses very clearly: the voice performance is still too long and not emotionally controlled enough, the dialogue needs better pacing and easier interruption, and the showcase should move into the central conflict earlier instead of spending too much time on setup. Another repeated point was that the presentation sometimes felt closer to a technical demo or trailer than to a clearly readable gameplay demonstration. For the final submission, these comments are useful because they point directly to the next design adjustments rather than rejecting the project direction itself.